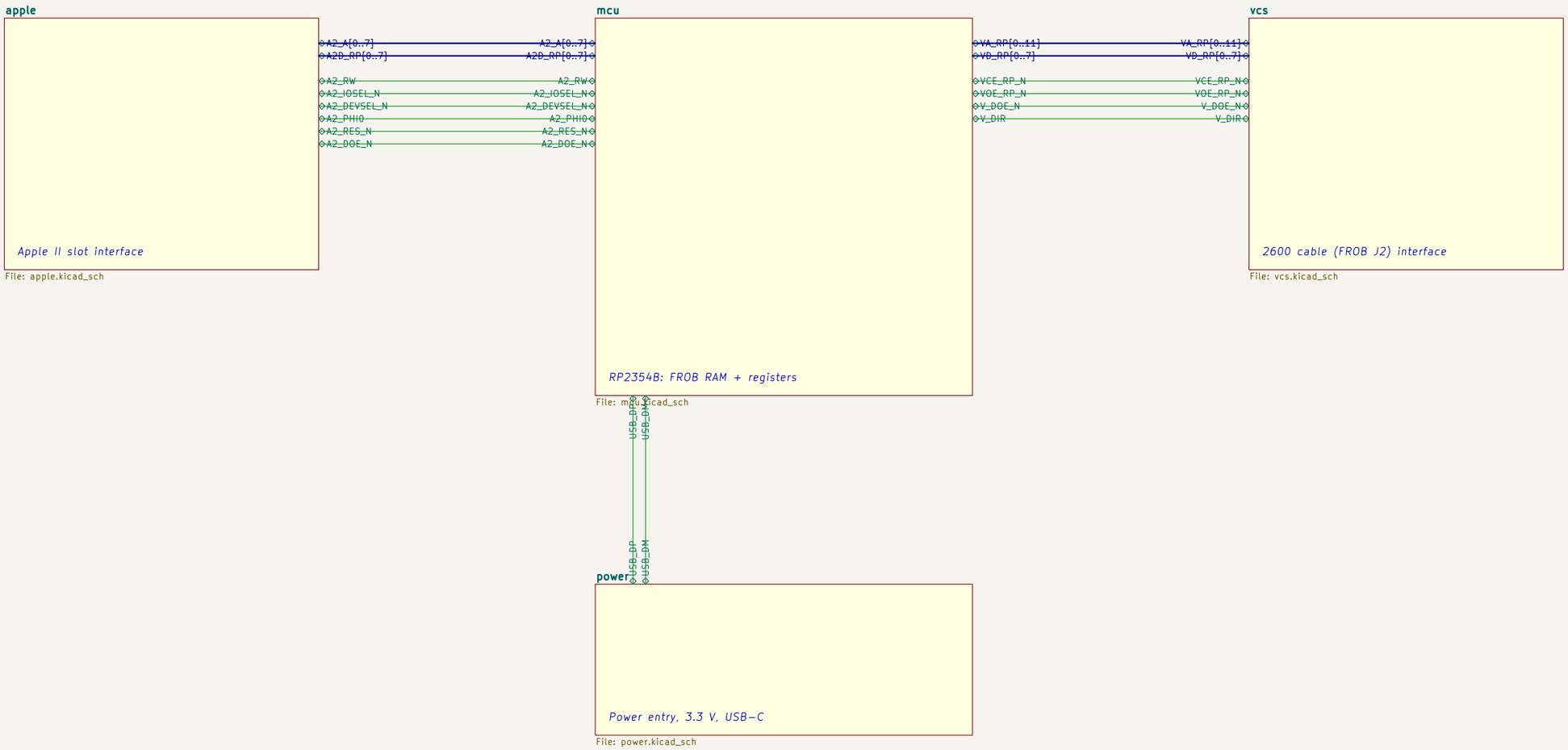
Frob2354 — the FROB Apple II card (Frobco, 1982) on one RP2354B

Atari 2600 development: 4 KB of RAM the Apple loads through a paged window ($Cn00–$CnFF) and the 2600 runs as cartridge ROM, plus a one-byte mailbox each way. Register-compatible with the original (see firmware/README.md); J2 takes the original FROB cable and cartridge adapter.



Apple $C0n0 write: D0–D3 page, D4 RAM to the 2600, D5 port on. Read: D7 may write, D6 may read.
Apple $C0n1: mailbox. 2600 $FFF0 write / $FFF1 status / $FFF2 read (port on).

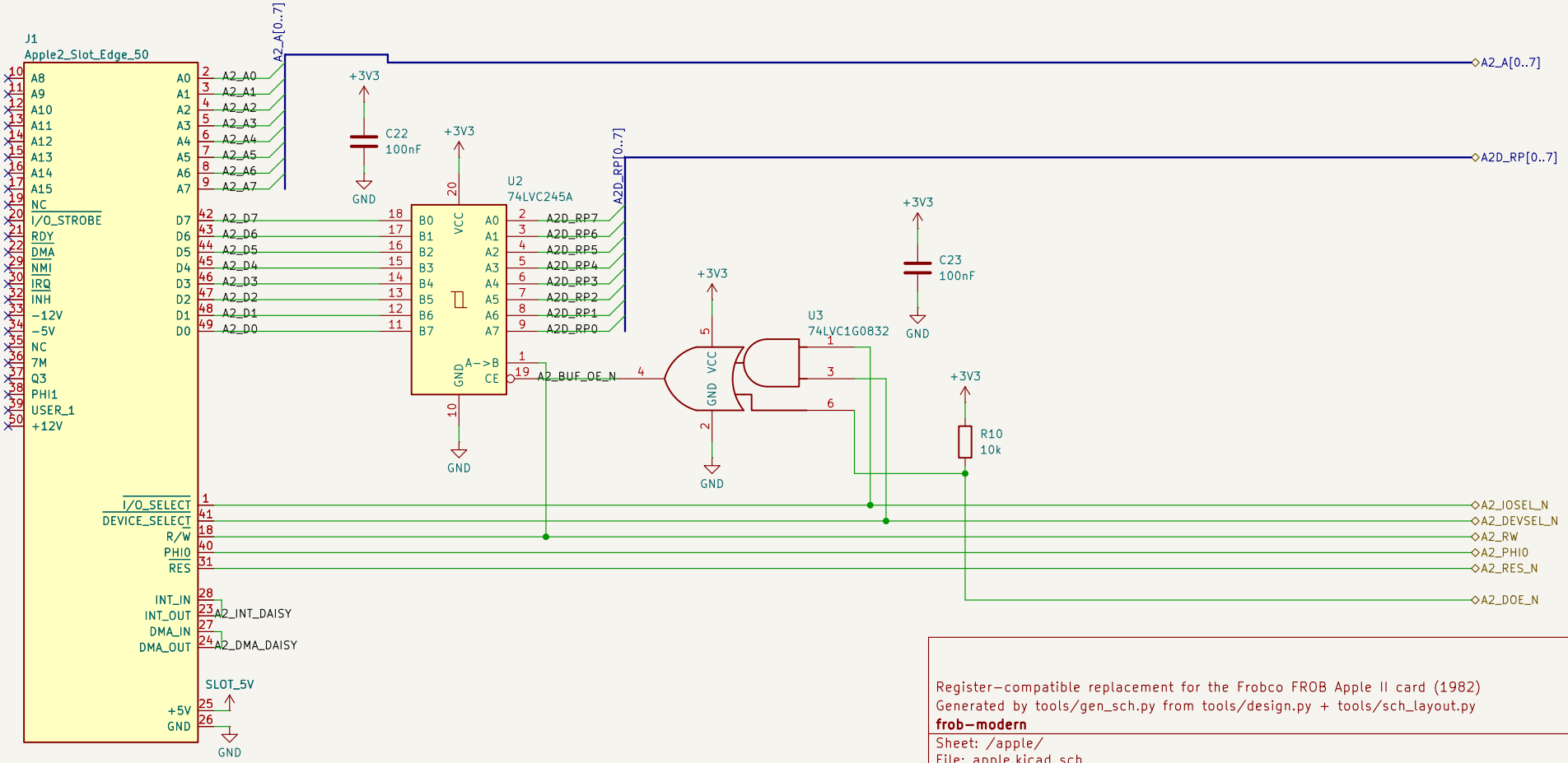
Generated by tools/gen_sch.py
frob-modern

Sheet: /
File: Frob2354.kicad_sch

Title: Frob2354

Size: A3	Date: 2026-10-02	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 1/5

Apple II slot interface
A0–A7, R/W, /IOSEL, /DEVSEL, PHI0, /RES go to the RP2354B directly (5 V–tolerant pads, GP8–20).
D0–D7 pass through U2 (74LVC245A): DIR = R/W, so a 6502 read drives the slot.
U2 /OE = (/IOSEL & /DEVSEL) | A2_DOE_N (U3): it can only drive during this slot's own cycles.
INT / DMA daisy chains passed through; unused slot pins (A8–A15, IRQ, NMI, ...) left open.



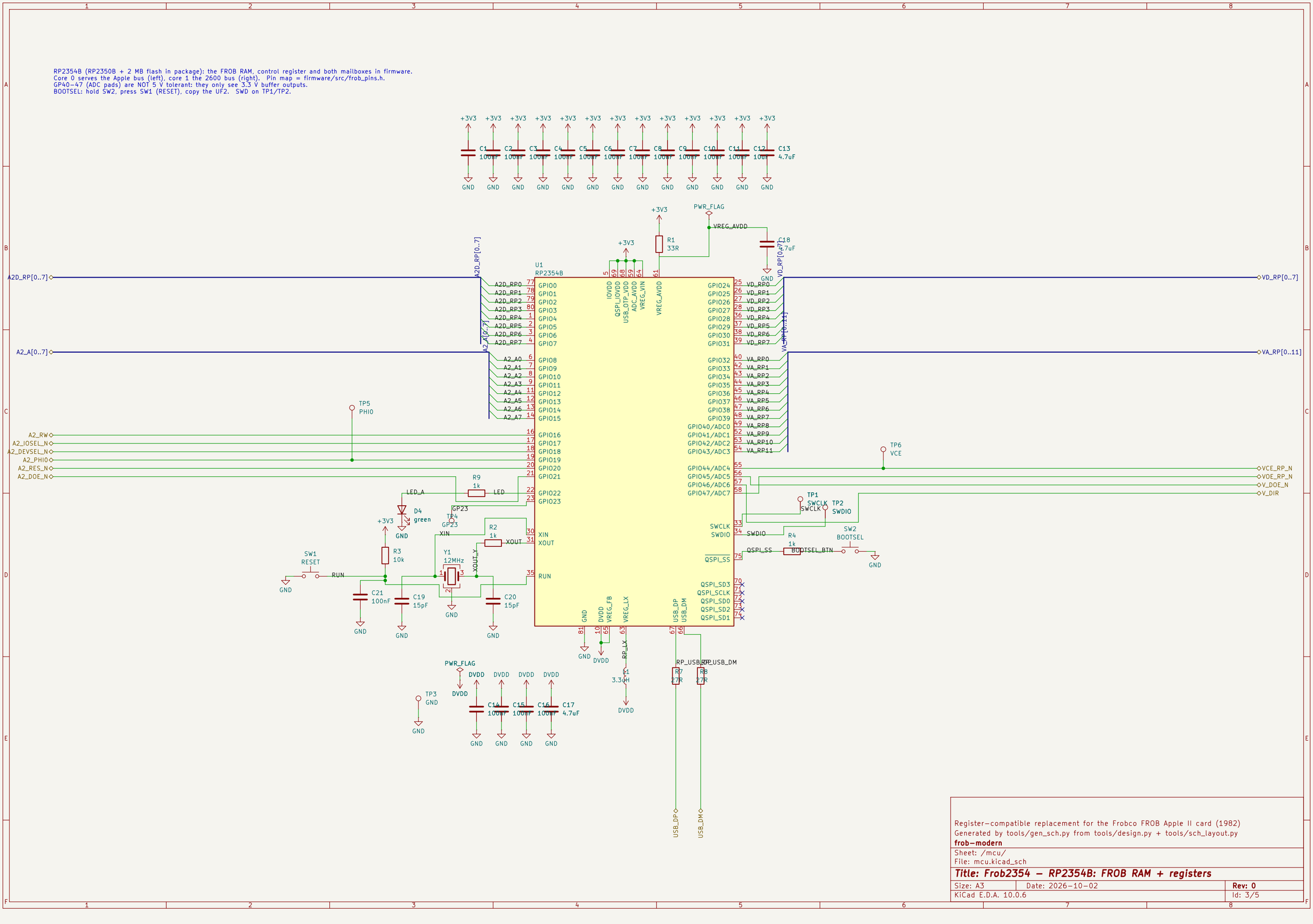
Register–compatible replacement for the Frobco FROB Apple II card (1982)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py
frob–modern

Sheet: /apple/
File: apple.kicad_sch

Title: Frob2354 – Apple II slot interface

Size: A4	Date: 2026–10–02	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 2/5

RP2354B (RP2350B + 2 MB flash in package): the FROB RAM, control register and both mailboxes in firmware.
Core 0 serves the Apple bus (left), core 1 the 2600 bus (right). Pin map = firmware/src/frob_pins.h.
GP40–47 (ADC pads) are NOT 5 V tolerant: they only see 3.3 V buffer outputs.
BOOTSEL: hold SW2, press SW1 (RESET), copy the UF2. SWD on TP1/TP2.



Register-compatible replacement for the Frobco FROB Apple II card (1982)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py
frob-modern

Sheet: /mcu/
File: mcu.kicad_sch

Title: Frob2354 — RP2354B: FROB RAM + registers

Size: A3	Date: 2026-10-02	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 3/5

2600 cable interface -- J2 is the original FROB card's DIP-24 socket (2732 pin-out; /CE = A12 inverted on the cable adapter, /OE grounded there, pin 24 open). Every cable signal goes through a 74LVC245A on the card's 3.3 V (5 V-tolerant, Ioff: the 2600 has its own supply). U7 /OE = /CE | V_DOE_N (U8).

VA_RP[0..11]

VD_RP[0..7]

VCE_RP_N

VOE_RP_N

V_DIR

V_DOE_N

VCS_CE_N

VBUF_OE_N

U5 74LVC245A

U6 74LVC245A

U7 74LVC245A

U8 74LVC1G32

J2 FROB cable

C29 100nF

C30 100nF

C31 100nF

C32 100nF

R12 10k

R13 10k

R14 10k

RN1 4k7

RN2 4k7

RN3 4k7

RN4 4k7

RN5 4k7

RN6 4k7

+3V3

GND

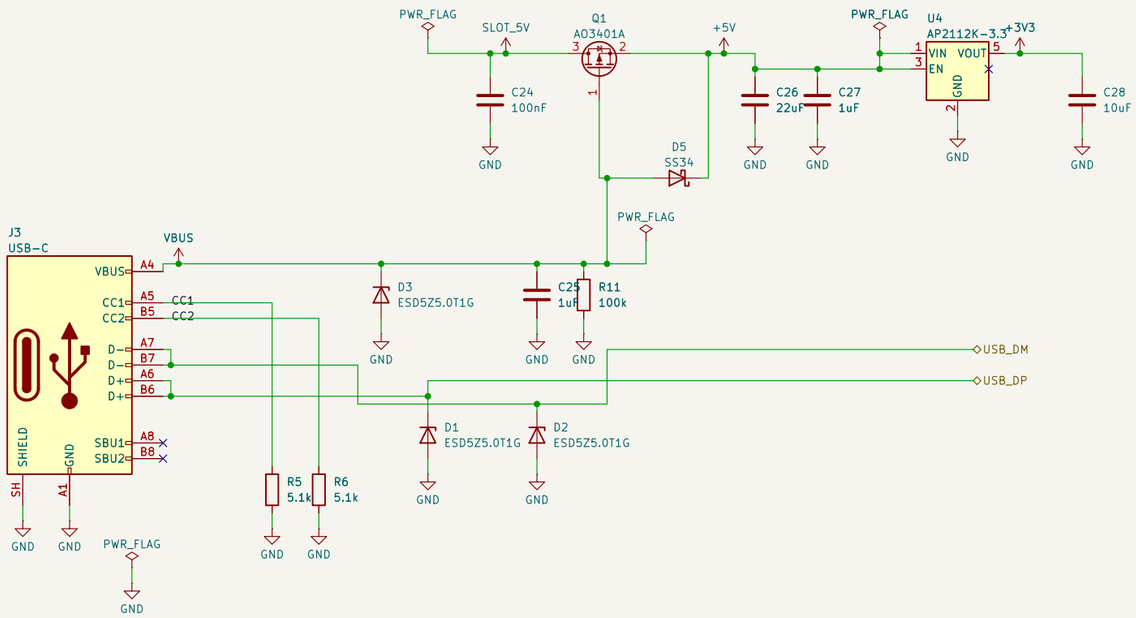
Register-compatible replacement for the Frobco FROB Apple II card (1982)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py
frob-modern

Sheet: /vcs/
File: vcs.kicad_sch

Title: Frob2354 — 2600 cable (FROB J2) interface

Size: A3	Date: 2026-10-02	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 4/5

Power entry. Slot +5V reaches +5V through Q1 (P-FET, gate = VBUS): fully on without USB, off with USB plugged in, so USB never back-feeds the Apple. USB VBUS reaches +5V through D5 (Schottky).
U4 (AP2112K, fast start) makes +3V3 for everything: the RP2354B pads are 5 V tolerant only while IOVDD is up, and the LDO tracks the slot rail as it rises. USB-C is for firmware updates (BOOTSEL).



Register-compatible replacement for the Frobco FROB Apple II card (1982)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py
frob-modern

Sheet: /power/
File: power.kicad_sch

Title: Frob2354 — Power entry, 3.3 V, USB-C

Size: A3	Date: 2026-10-02	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 5/5